

Electricity and Electrical Safety

Programmable Logic Controllers

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College of Information Technology and Engineering

Agenda

Electricity Review

Circuit Review

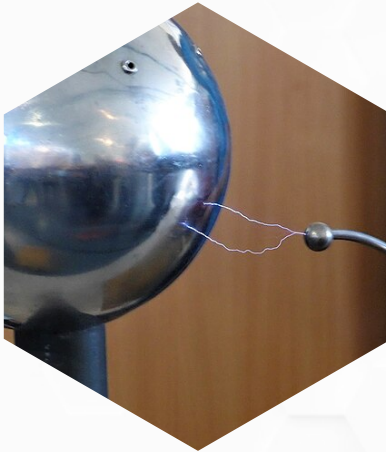
Electrical Safety

Student Learning Outcomes

1. Create programs in Ladder Logic using the following instructions in an application
2. Program PLC timers and counters

Electricity Review

Electricity



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Electricity is a form of energy resulting from the existence of charged particles (protons or electrons)

Primary Properties of Electricity

1. Voltage
2. Current
3. Resistance



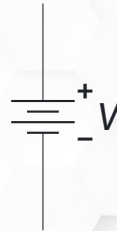
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Electricity Review

Voltage

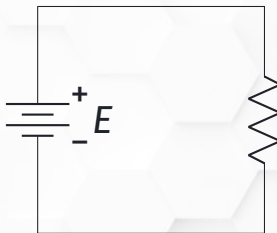
Voltage as Potential Energy

- Voltage is potential energy
- The difference between the number of electrons in the positive and negative terminals
- Potential energy + difference = *potential difference*



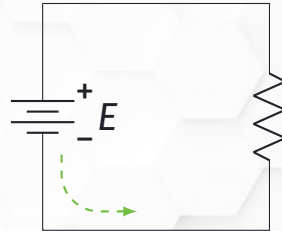
Voltage as Electromotive Force

- Voltage acts like a pressure—on its own it's a pressurized tank
- When connected to a circuit it propels electrons through the circuit
- *Electromotive force (EMF)*



Voltage Generates Current

- Voltage pushed electrons around the circuit, creating *current*



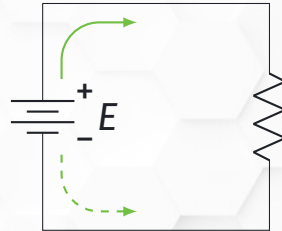
Electron Flow

- Law of physics: things move from high concentration to low concentration
- Current flows from high concentration of electrons (negative) to low concentration (positive)



Conventional Flow

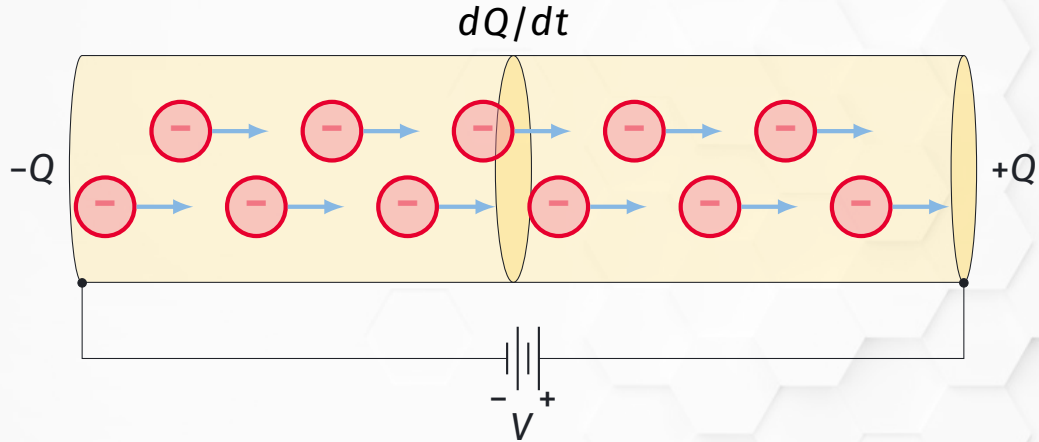
- Current is traditionally drawn *flowing from positive to negative*
- *Passive sign convention*
- Mathematically fine as long as directions and polarities are always maintained



Electricity Review

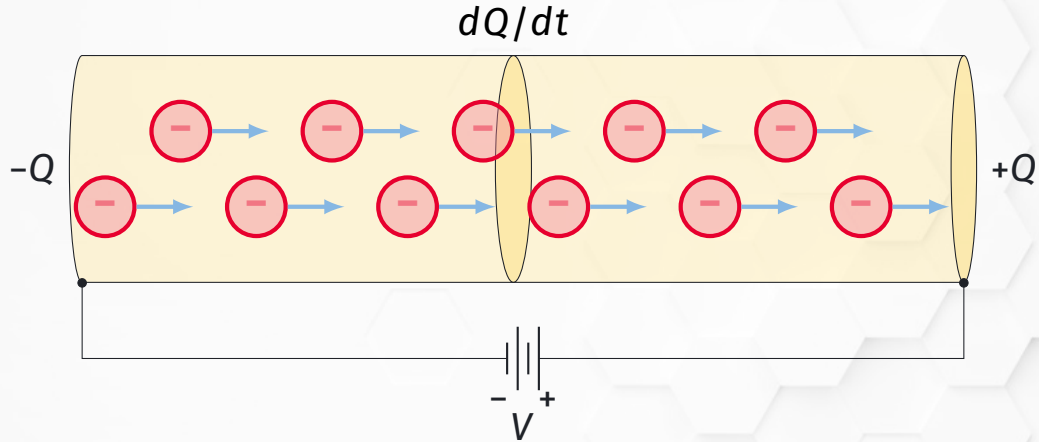
Current

Current Defined



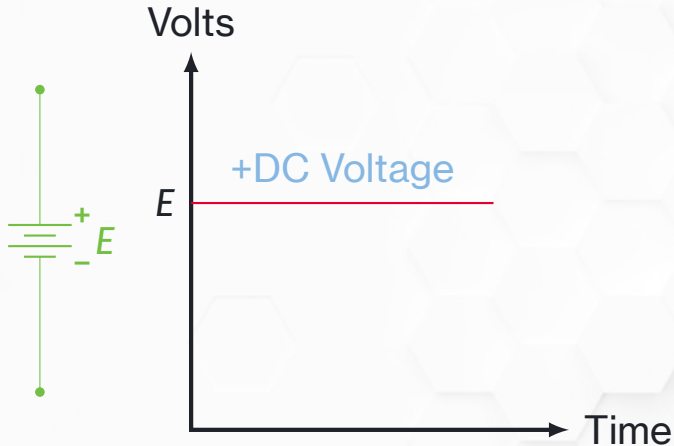
Current is the *flow rate* (speed) of electric charge through a conductor

Ampere Defined



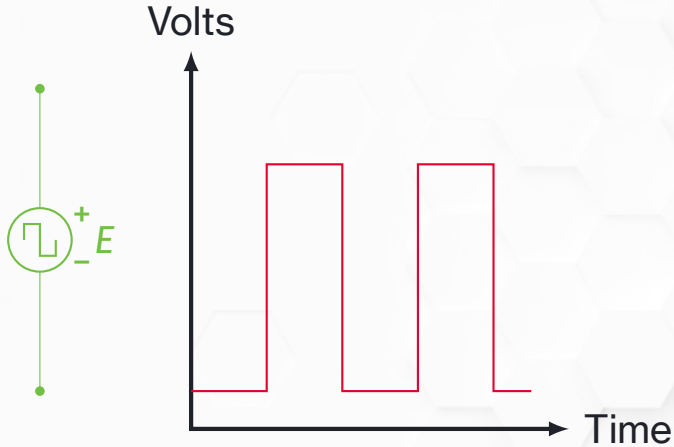
One *ampere* is 1 coulomb of charge (6.24×10^{18} electrons) flowing past a single point in 1 second

Direct Current Defined



Direct current is an electric current that maintains its polarity (flows in only one direction)

Pulsed DC



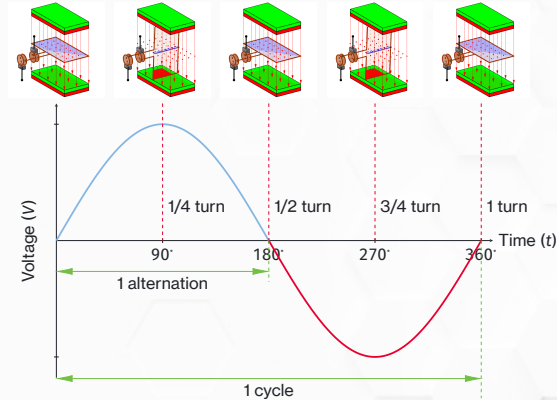
Direct current **does not** need to be a steady voltage – it can be *pulsed*. As long as it maintains polarity it is still DC.

Alternating Current Defined



Alternating current is an electric current that reverses polarity. It **does not** need to be a sine wave, but the sine wave is the most common.

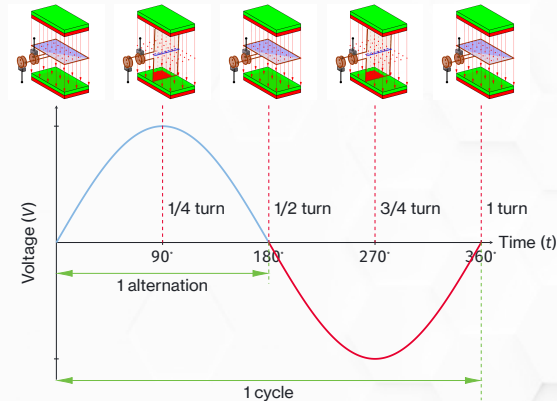
AC Voltage - Sine Wave



Attribution: Adapted from "Electric Generator Animation" (frames 5 and 17 extracted) by MikeRun, [CC BY-SA 4.0](#), via Wikimedia Commons.

The sine wave is caused by the physical layout of an AC generator – as the coil spins inside the magnetic field it cuts the field in different directions, leading to a reverse in polarity

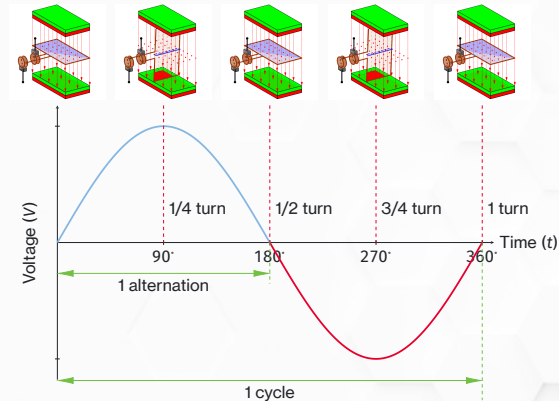
AC Voltage - Alternation



Attribution: Adapted from "Electric Generator Animation" (frames 5 and 17 extracted) by MikeRun, CC BY-SA 4.0, via Wikimedia Commons.

An alternation is 1/2 of a rotation of the coil—the point at which the polarity changes

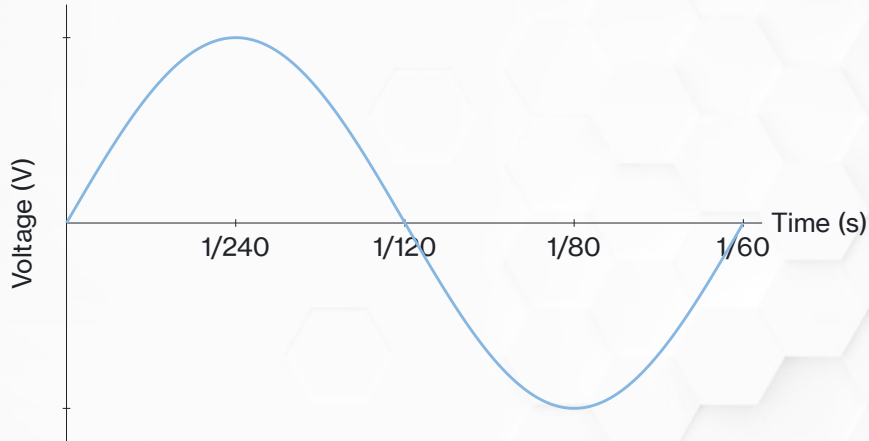
AC Voltage - Cycle



Attribution: Adapted from "Electric Generator Animation" (frames 5 and 17 extracted) by MikeRun, CC BY-SA 4.0, via Wikimedia Commons.

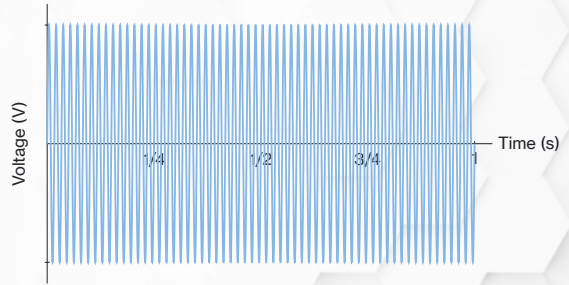
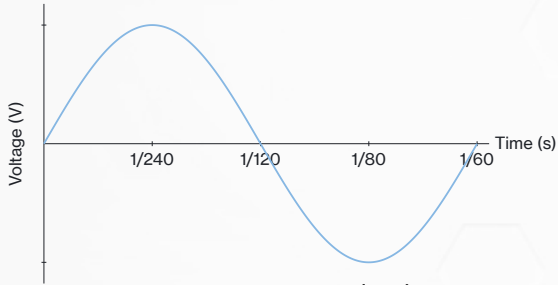
A cycle is a full rotation of the coil

AC Voltage - Frequency



The frequency is the number of cycles per second (Hz)–the U.S. standard is 60 Hz (60 cycles per second)

Visualizing Frequency

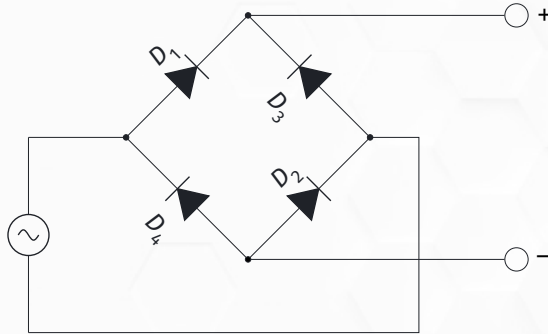


We often use one cycle (left) to visualize the curve, but remember that that one cycle is happening 60 times every second (right)

Electricity Review

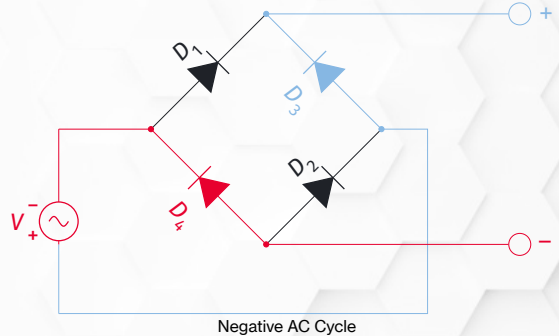
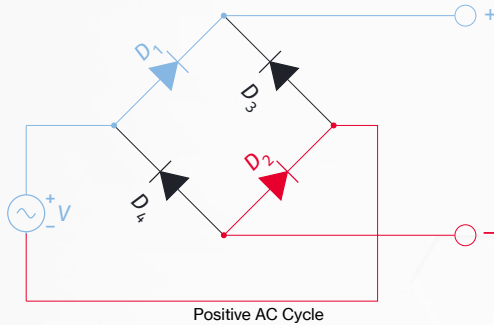
Converting AC to DC

Bridge Rectifier



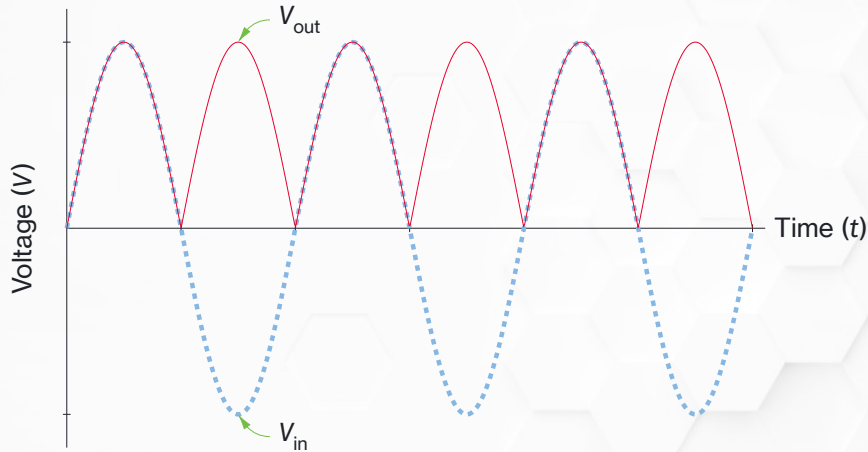
A bridge rectifier is commonly used in controls applications to convert AC to DC

Bridge Rectifier - Active



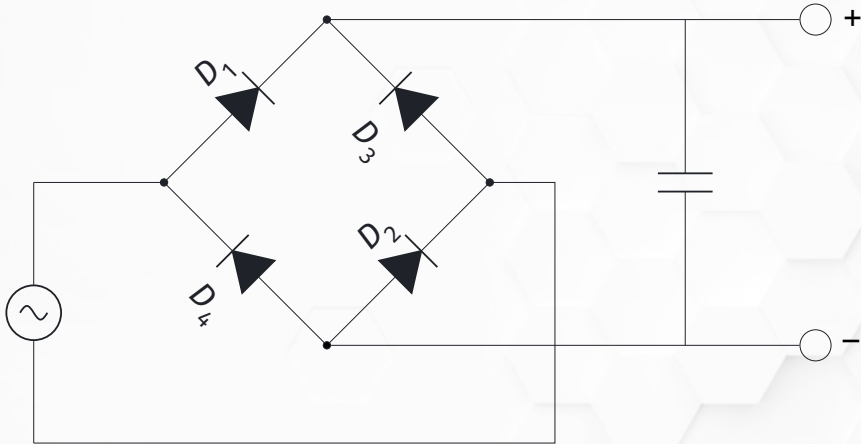
As the polarity of the AC voltages changes the the rectifier forces the output to the same polarity

Rectified Alternating Current



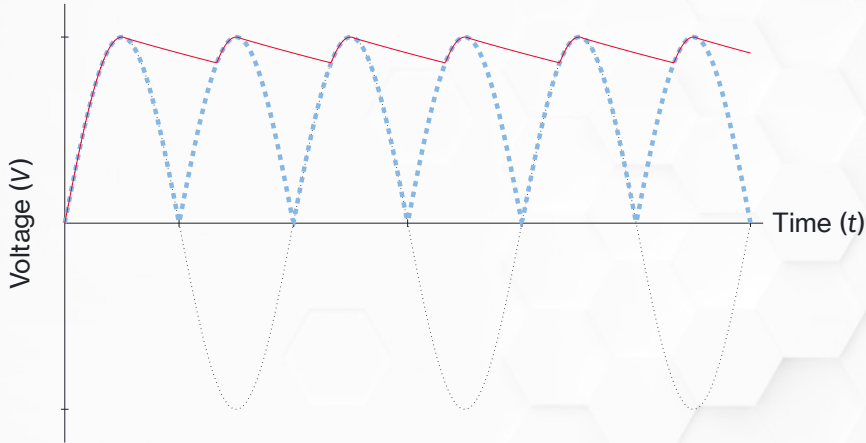
The rectifier redirects current so the output is always the same polarity, but it *does not* affect the voltage value—the resulting wave is DC, but not very useful DC

Bridge Rectifier - Smoothed



Smoothing capacitors are added to help maintain the voltage level between cycles, which helps keep the DC at a more consistent value

Rectified and Smoothed Alternating Current

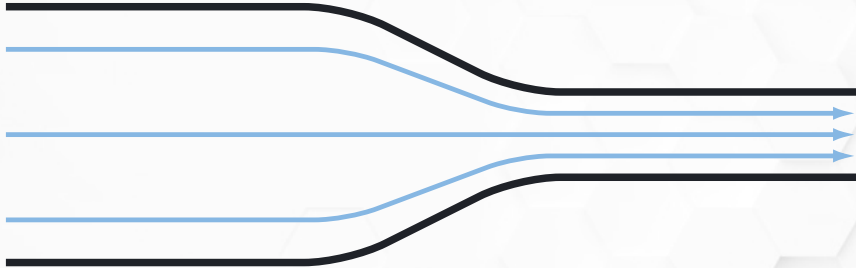


The result is a fairly smooth DC wave—the *ripple* will always be there, but can be tweaked by changing the capacitor value

Electricity Review

Resistance

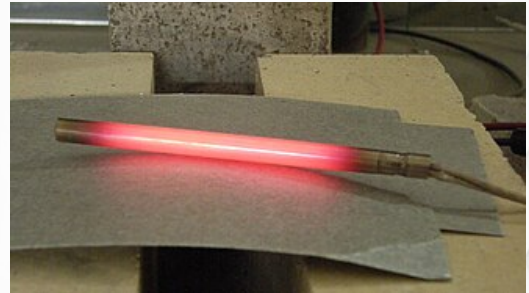
Resistance Defined



Resistance is the opposition to electron flow—like a restriction in a water pipe

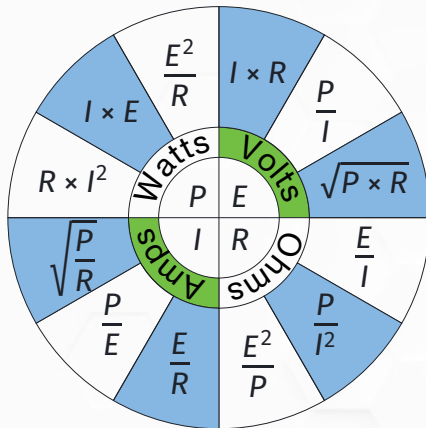
Resistance Causes Heat

- *ALL* components in a circuit have resistance—including wires
- Current through a resistance *always* comes with heat
- Unless we're actually building a heater, this is *dissipated* (lost) energy



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Ohm's Law

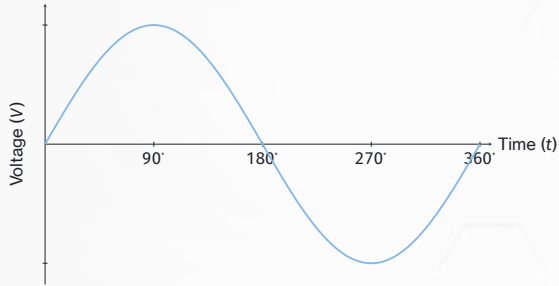


Ohm's Law relates current, voltage, and resistance. Combined with Watt's Law to relate power (heat) we can solve all aspects of a circuit.

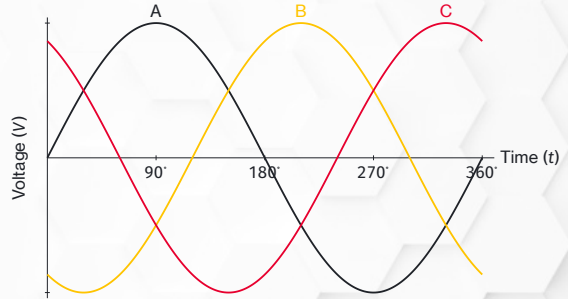
Electricity Review

Voltage Phases

AC Voltage - Phases

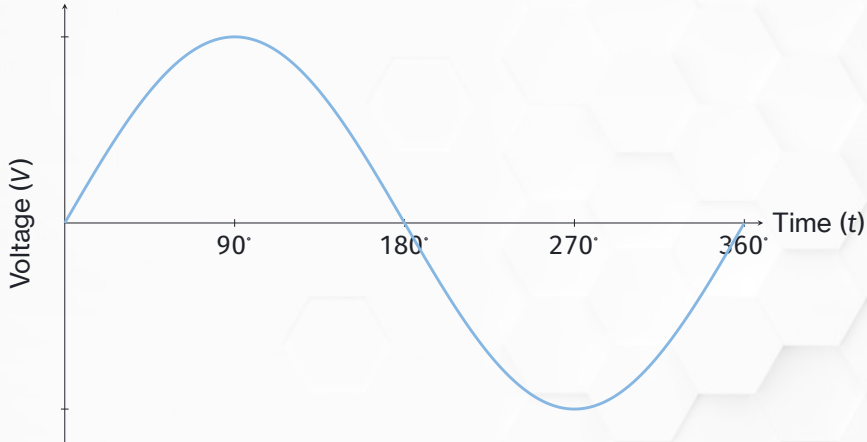


1 ϕ is created with a single alternator coil



3 ϕ is created with 3 coils physically placed 120° apart

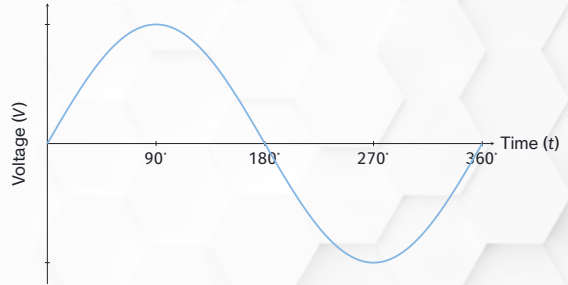
Single Phase - Zero Voltage (1 of 5)



The instantaneous voltage value of 1 ϕ crosses 0 twice every cycle

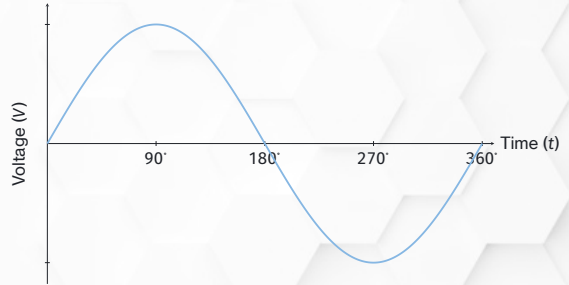
Single Phase - Zero Voltage (2 of 5)

- What happens when voltage drops to 0?



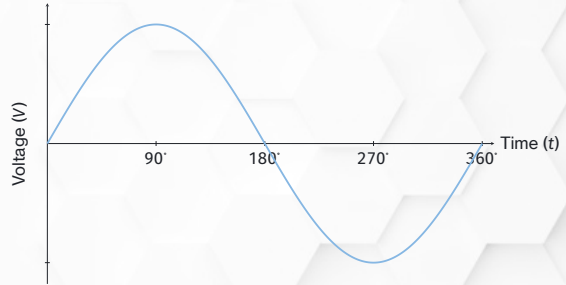
Single Phase - Zero Voltage (3 of 5)

$$p(t) = v(t) \cdot i(t)$$
$$v(t) = 0 \rightarrow p(t) = 0$$

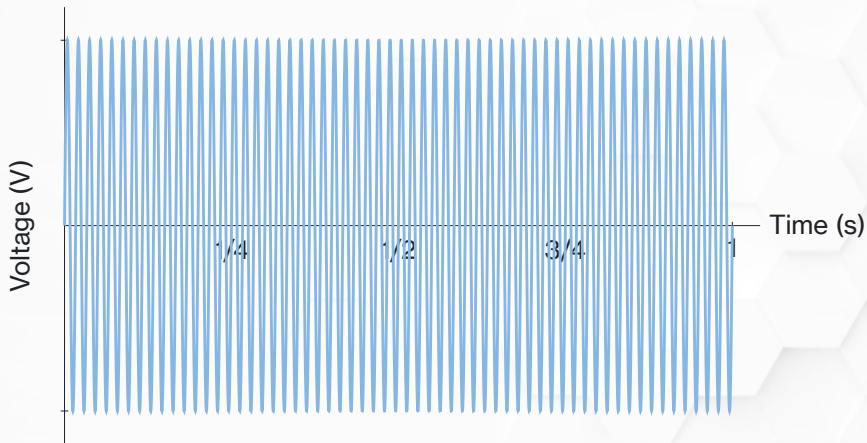


Single Phase - Zero Voltage (4 of 5)

- Instantaneous power drops to zero twice every cycle
- Requires higher peak current I_{peak} to deliver equal average power P_{avg}
- Causes mechanical vibration and pulsating torque in heavy equipment

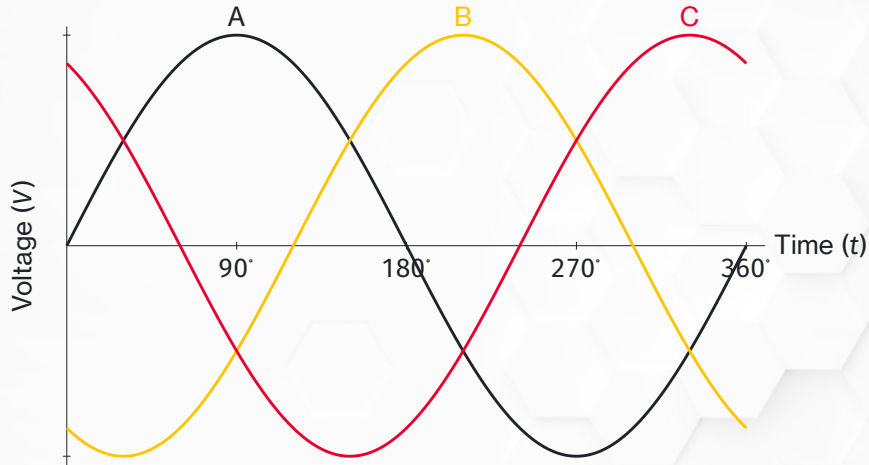


Single Phase - Zero Voltage (5 of 5)



This isn't a rare occurrence—at 60 Hz the power is dropping to zero 120 times every second

Three Phase Power



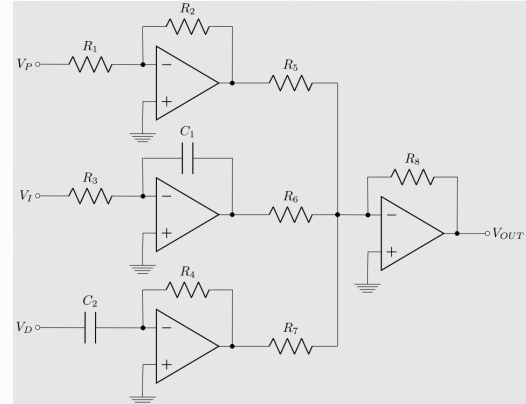
With a 3 ϕ power supply the effective voltage is *never* zero—less average current required to maintain power and no pulsing

Circuit Review



Electric Circuit Defined

- A closed, complete loop of conductive components that allow current to flow from a source, through a device, and back to the source

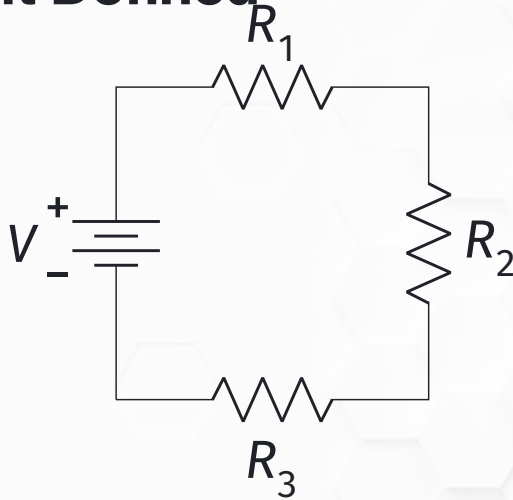


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Circuit Review

Series Circuits

Series Circuit Defined

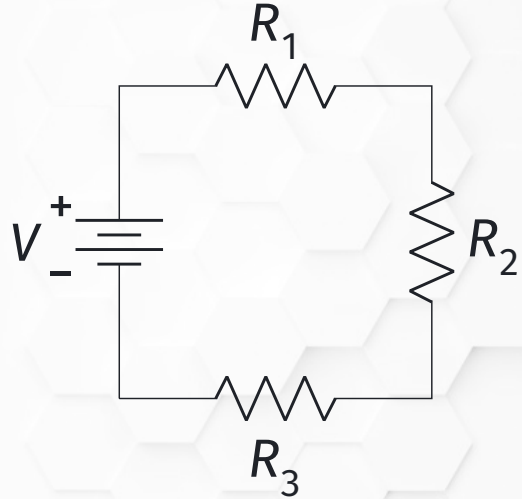


A *series circuit* is a circuit that has only one path for current to follow—if that path is broken, current stops

Resistance in Series

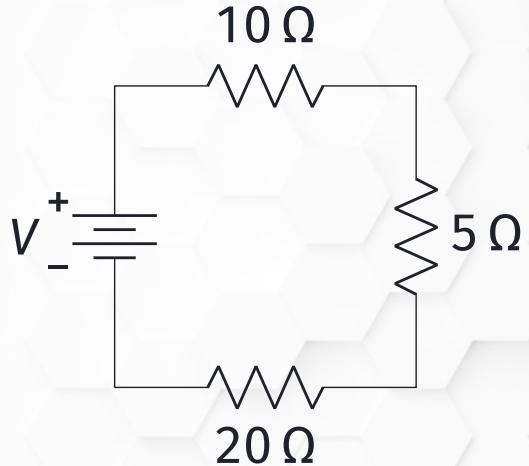
- Resistance in series *adds*

$$R_T = \sum_{i=1}^n R_i = R_1 + R_2 + \dots + R_n$$



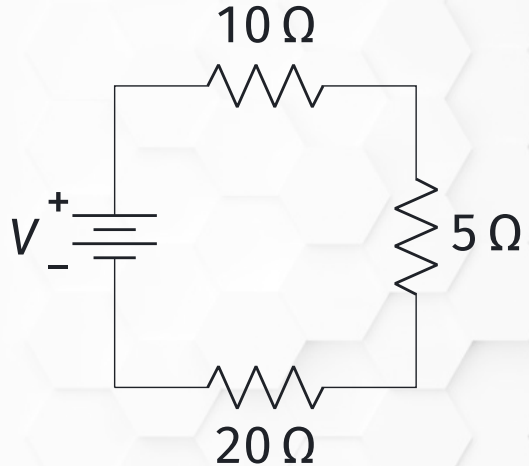
Resistance in Series Example 1

- What is the total resistance of this circuit?



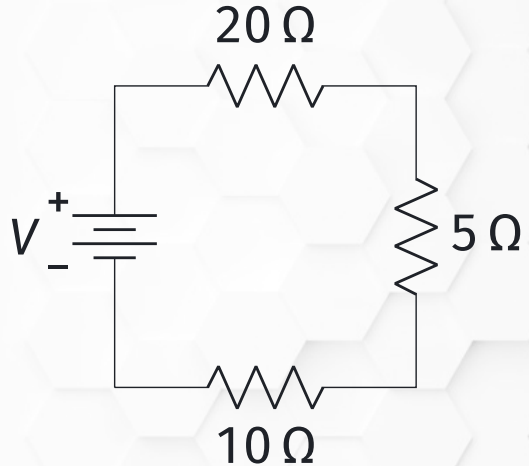
Resistance in Series Example 1 Solution

$$\begin{aligned} R_T &= \sum_{i=1}^3 R_i = R_1 + R_2 + R_3 \\ &= 10\ \Omega + 5\ \Omega + 20\ \Omega \\ &= 35\ \Omega \end{aligned}$$



Resistance in Series Example 2

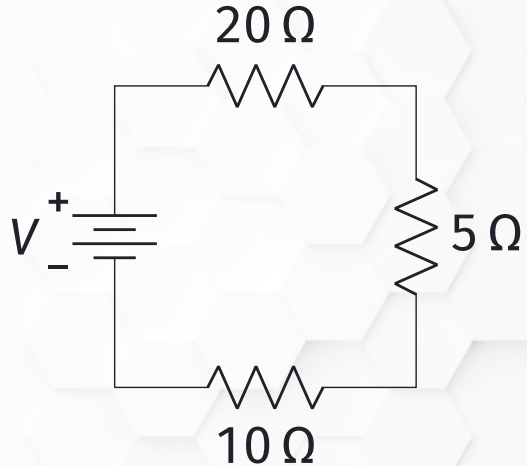
- What is the total resistance of this circuit?



Resistance in Series Example 2 Solution

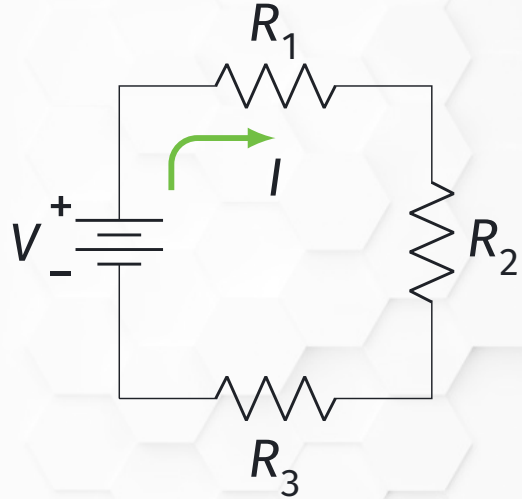
$$\begin{aligned} R_T &= \sum_{i=1}^3 R_i = R_1 + R_2 + R_3 \\ &= 20\ \Omega + 5\ \Omega + 10\ \Omega \\ &= 35\ \Omega \end{aligned}$$

- Order doesn't matter...



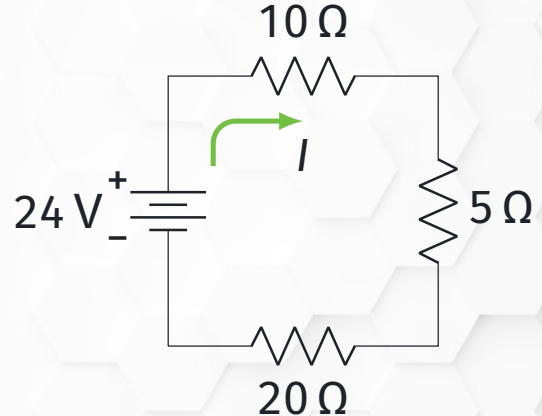
Current in Series

- One path for current = current is equal at all points
- Ohm's Law – current in a series circuit is equal to the voltage divided by the circuit total resistance



Current in Series Example

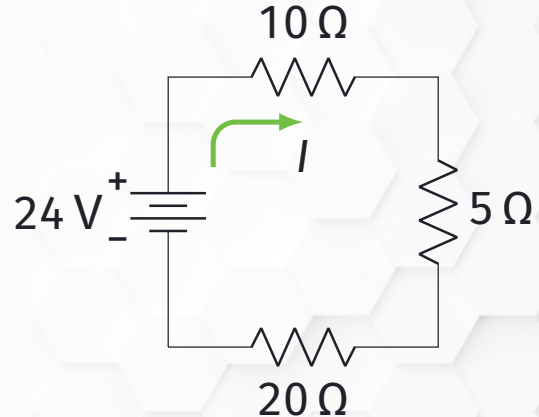
- What is the total current draw of this circuit?



Current in Series Example Solution

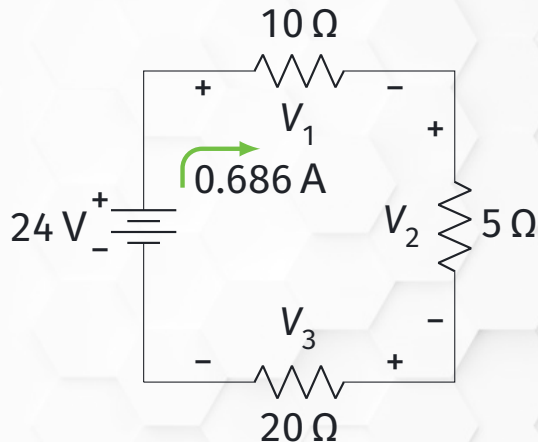
$$R_T = 35 \Omega$$

$$I = \frac{E}{R} = \frac{24 \text{ V}}{35 \Omega} = 0.686 \text{ A}$$



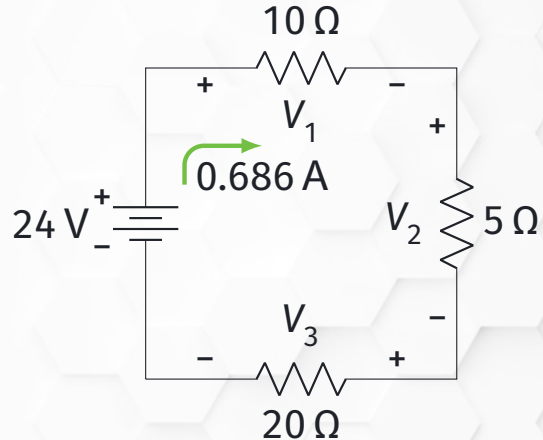
Voltage in Series

- Resistors work by robbing electrons of their energy
- The energy is voltage, this loss in energy causes a *voltage drop* across each resistor
- Ohm's Law can be used to calculate the drop



Voltage in Series Example

- What is the voltage drop across each of these resistors?



Voltage in Series Example Solution

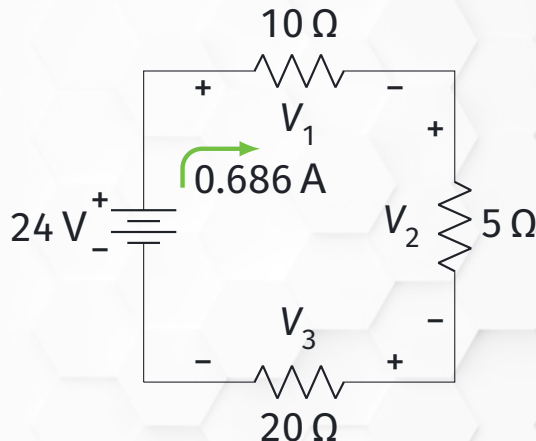
$$E = IR$$

$$V_1 = IR_1 = (0.686 \text{ A})(10 \Omega) = 6.86 \text{ V}$$

$$V_2 = IR_2 = (0.686 \text{ A})(5 \Omega) = 3.43 \text{ V}$$

$$V_3 = IR_3 = (0.686 \text{ A})(20 \Omega) = 13.72 \text{ V}$$

$$V_T = 6.86 \text{ V} + 3.43 \text{ V} + 13.72 \text{ V} = 24 \text{ V}$$

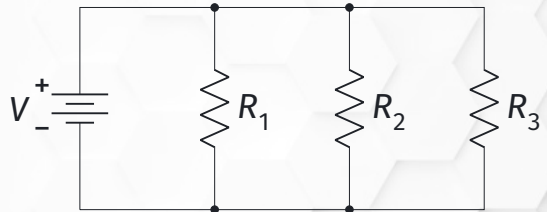


Circuit Review

Parallel Circuits

Parallel Circuits

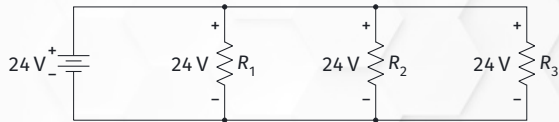
- Two or more components connected such that there is more than one path for current to take
- De-energizing one path may leave other paths energized



Voltage in Parallel

- The voltage across any parallel component is equal to the voltage across all other parallel components
- In more complex circuits the V is the voltage in a parallel *leg*, not the supply

$$V = V_1 = V_2 = V_3$$

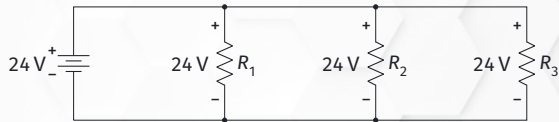


Resistance in Parallel

- Resistance in parallel *does not* add, *conductance* does

$$G = \frac{I}{E} = \frac{1}{R}$$

$$\frac{1}{R_{EQ}} = \sum_{i=1}^n \frac{1}{R_i} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}$$



Resistance in Parallel Example

- What is the equivalent resistance of this circuit?



Resistance in Parallel Example Solution

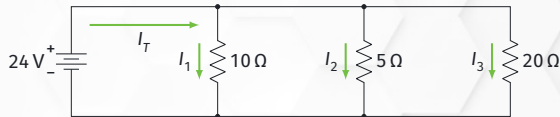
$$\begin{aligned}\frac{1}{R_{EQ}} &= \sum_{i=1}^3 \frac{1}{R_i} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \\ &= \frac{1}{10\ \Omega} + \frac{1}{5\ \Omega} + \frac{1}{20\ \Omega} \\ &= \frac{7}{20\ \Omega} \\ R_{EQ} &= \frac{20\ \Omega}{7} = 2.857\ \Omega\end{aligned}$$



Sanity check: the equivalent resistance of a parallel branch will *a/ways* be lower than the smallest individual resistance

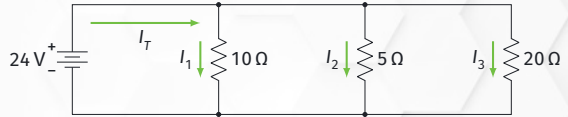
Current in Parallel

- Remember the definition: parallel circuits have multiple current paths
- The current splits every time there's a junction
- *How much* it splits is determined by Ohm's Law



Current in Parallel Example

- What is the current through each resistor in this circuit?

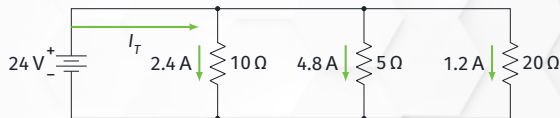


Current in Parallel Example Solution

$$I_1 = \frac{E}{R_1} = \frac{24 \text{ V}}{10 \Omega} = 2.4 \text{ A}$$

$$I_2 = \frac{E}{R_2} = \frac{24 \text{ V}}{5 \Omega} = 4.8 \text{ A}$$

$$I_3 = \frac{E}{R_3} = \frac{24 \text{ V}}{20 \Omega} = 1.2 \text{ A}$$



Electrical Safety



Electric Shock

Current Vs Voltage: How Much Current Can Kill You? (5:21)

https://www.youtube.com/watch?v=y_cWTWB-N_I

Arc Flash

Electrical Arc Flash Demonstration (1:17)

<https://www.youtube.com/watch?v=6hpE5LYj-CY>

Electrical Safety

Electrical Standards

National Fire Protection Association

- Began as a safety organization focused on preventing fires in the workplace
- Currently publishes several important electrical standards



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NFPA Standards

- NFPA 70 – National Electric Code (NEC)
- NFPA 70e – Standard for Electrical Safety in the Workplace
- NFPA 79 – Electrical Standards for Industrial Machinery



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- Hard copies of these standards can be expensive
- **NFPA's website** offers free viewing of all standards
- **HIGHLY** recommend creating a free account



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Electrical Safety

Electrical Enclosures

Electrical Enclosures

- Enclosures are selected based on their use scenario
- Two standards for rating enclosures:
 - National Electronics Manufacturers Association (NEMA)
 - International Electrotechnical Commission (IEC)



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NEMA Enclosure Ratings (1 of 4)

- ALL** ratings protect personnel from hazardous parts
- It's a box – you can't touch the stuff inside it



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NEMA Enclosure Ratings (2 of 4)

- R** added to rating means it includes a drain hole for water
- X** added to rating means it is corrosion resistant (stainless, plastic, fiberglass, etc.)



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NEMA Enclosure Ratings (3 of 4)

NEMA 1

- Indoor use only
- Protects equipment from falling dust

NEMA 3

- Indoor/outdoor
- Protects from falling and windblown dust
- Protects from incidental weather



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NEMA Enclosure Ratings (4 of 4)

- Indoor/outdoor
- Protects from incidental and hose directed water



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IEC Enclosure Ratings

- Ingress Protection (IP) rating
- Defines sealing effectiveness against foreign bodies (solids) and moisture
 - First digit = intrusion protection (solids)
 - Second digit = moisture protection
- X means the enclosure is not rated for that spec



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IP Ratings – Intrusion Protection

- 0** No protection
- 1** >50mm diameter (hands)
- 2** <80mm length, >12mm diameter (fingers)
- 3** >2.5mm diameter (tools, wires)
- 4** >1mm diameter (wires, nails, etc.)
- 5** Partial protection against dust
- 6** Totally dust tight

IP Ratings – Moisture Protection

- 0** No protection
- 1** Vertically falling droplets
- 2** Droplets up to 15° from vertical
- 3** Spray up to 60° from vertical
- 4** Splashes from all directions
- 5** Low pressure jets, all directions
- 6** Direct, high pressure jets
- 7** Full immersion up to 30 min at 1m
- 8** Extended immersion at high pressure
- 9** High pressure, high temperature jets

Electrical Safety

Grounding and Bonding

Grounding

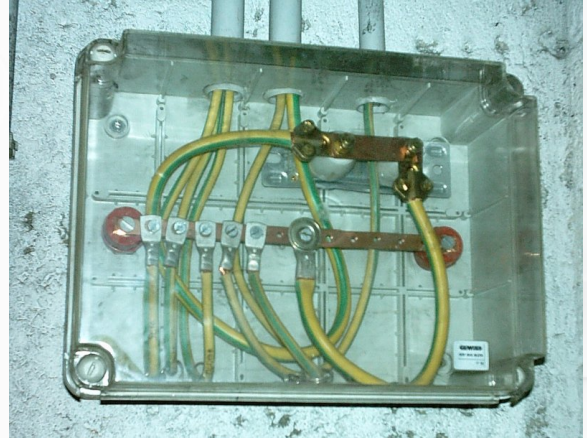
- Grounding – the connection of all non-current carrying components to the Earth
- Earth is a giant electrical reservoir
 - Readily absorbs stray electricity
 - No stray electricity = no electrical shock



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Bonding

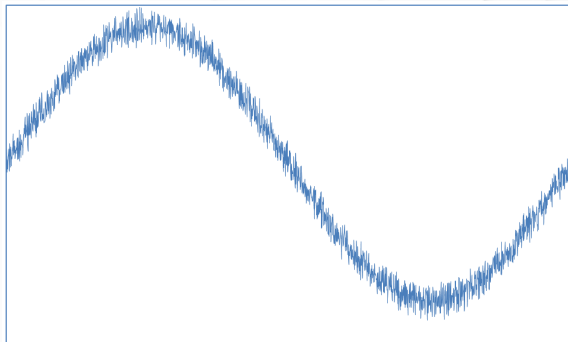
- Bonding – the connection of two separate non-current carrying components to one another
- Grounding cannot always be used
- Reduces risk of shock by keeping both components at same potential



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Electrical Noise

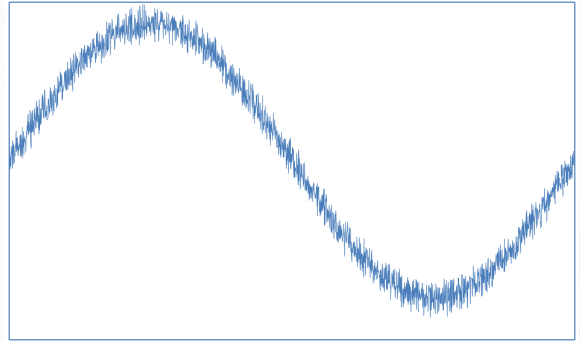
- Any unwanted signal on an electrical component
- Not just power wires—communication wires are *especially* noise prone



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Mitigating Noise

- PLCs should be placed away from noise generating devices
 - Motors, starters, welders, etc.
- AC and DC conductors should be separated
 - If they must cross, **ALWAYS** cross at 90°



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Noise and Safety (1 of 2)

Why is noise in the safety section of this presentation?

Noise and Safety (2 of 2)

If the noise is severe enough it can cause false signals to the PLC—could cause a machine to move when it's not supposed to

Electrical Safety

Personal Protective Equipment

Protective Clothing

- NFPA 70e requires employees to wear flame resistant (FR) clothing whenever exposure to arc flash is possible



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FR Clothing

- Must not continue to burn when ignited
- Must insulate well enough to dissipate heat away from skin
- Must resist the break open force of the shockwave



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Eye Protection

- OSHA 29 CFR 1910.133 Eye and Face Protection
- Safety glasses
- Goggles
- Face shield
- Arc flash hood



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Hand Protection

- Live electrical work requires the use of rubber insulating gloves
- Leather gloves are worn **OVER** rubber gloves to protect the rubber
- Leather gloves **ARE NOT** a sufficient insulator alone



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Personal Protective Equipment

PPE SAFETY VIDEO | Testing common types of PPE (3:48)

https://www.youtube.com/watch?v=nw_pNMnLG3A

Electrical Safety

Lockout/Tagout

Control of Hazardous Energy

- OSHA 29 CFR 1910.147
Control of Hazardous Energy
 - Note: **ENERGY** – LOTO is not limited to electricity
- LOTO should be carried out whenever any piece of equipment is being serviced, repaired, or replaced



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Lockout/Tagout

1. Prepare for shutdown
2. Shut down
3. Isolate power sources
4. LOTO power sources
5. Release stored energy
6. Verify isolation



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Why LOTO Is So Important

First Day on the Job Was His Last: What Happened to Day Davis
(2:50)

<https://www.youtube.com/watch?v=cNcsTRQNZLE>

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